

10 (a) Fig. 10.1 shows chemoreceptor cells in a taste bud. Two of the chemoreceptor cells have formed synapses with sensory neurone dendrites.

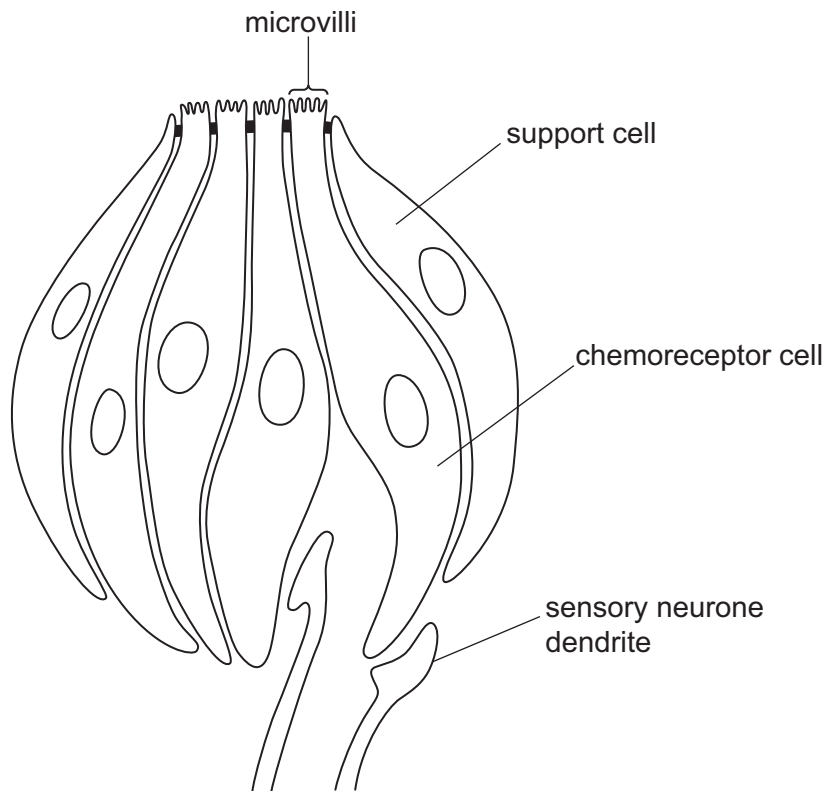


Fig. 10.1

Describe how the contact of sodium ions with the microvilli of the chemoreceptor cell can lead to the release of a neurotransmitter by the cell.

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- (b) Some of the neurotransmitters in the brain are produced through a series of reactions (reaction pathway) from a chemical called DOPA.

DOPA is also involved in other reaction pathways. For example, in the skin and eyes, DOPA is part of a different reaction pathway that depends on the *TYR* gene.

Describe **and** explain the phenotypic consequences for the skin and eyes of a person who is homozygous for a mutated, non-functional allele of the *TYR* gene.

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